Math 242, S20 **Name:** Answer

Quiz 8

<u>**Instructions:**</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (6 points) Find the general solution of the given nonhomogeneous differential equation:

$$y'' - 2y' - 3y = \cos x.$$

Solution. step 1. Find the complementary solution $y_c(x)$.

characteristic equation:

$$r^2 - 2r - 3 = 0$$
$$(r-3)(r+1) = 0$$
$$\Rightarrow r_1 = 3,\ r_2 = -1$$
$$\Rightarrow y_c(x) = C_1 e^{3x} + C_2 e^{-x}$$

step 2. Find a particular solution $y_p(x)$

$$f(x) = \cos x,\ f'(x) = -\sin x,\ f''(x) = -\cos x,$$
$$f^{(3)}(x) = \sin x,\ f^{(4)}(x) = \cos x,\ \cdots$$

Thus we have the trial function

$$y_p(x) = A\cos x + B\sin x$$

$$y_p'(x) = -A\sin x + B\cos x$$
$$y_p''(x) = -A\cos x - B\sin x.$$

Put them into the homogeneous equation

$$y_p'' - 2y_p' - 3y_p = (-A\cos x - B\sin x) - 2(-A\sin x + B\cos x) - 3(A\cos x + B\sin x)$$
$$= \underline{(-4A-2B)}\cos x + \underline{(2A-4B)}\sin x$$
$$\equiv \cos x$$

$$\Rightarrow \begin{cases} -4A - 2B = 1 \\ 2A - 4B = 0 \end{cases} \Rightarrow \begin{array}{l} A = -\frac{1}{5} \\ B = -\frac{1}{10} \end{array}$$

$$\Rightarrow y_p = -\frac{1}{5}\cos x - \frac{1}{10}\sin x$$

$$\Rightarrow y(x) = y_c(x) + y_p(x)$$
$$= C_1 e^{3x} + C_2 e^{-2x} - \frac{1}{5}\cos x - \frac{1}{10}\sin x$$

Problem 2. (4 points) Determine the appropriate form for a particular solution y_p of the nonhomogeneous differential equation:

$$y'' - 2y' - 3y = xe^{-x}.$$

Solution. From problem 1, we know

$$y_c(x) = C_1 e^{3x} + C_2 e^{-x}$$

The righ-hand side function $f(x) = xe^{-x}$, thus

$$y_p(x) = x^s(Ax+B)e^{-x}$$

avoid duplication of terms of $f(x)$ in $y_c(x)$.

$$\overset{s=1}{\equiv} x(Ax+B)e^{-x}$$
$$= (Ax^2 + Bx)e^{-x}$$